

Booking Dashboard

Logic & Calculation Document

LabWise LIMS

8 April 2026

Stats Strip (Top Bar)

Booked Today

Count of Booking records where bookedBy = this user AND createdAt = today. Shows % trend vs yesterday.

Pending Queue

Global count of Sample records where status = "received". Lab-wide metric, not user-scoped.

Revenue

SUM(Booking.netAmount) where bookedBy = this user AND createdAt = today. Shown in thousands.

On-Time Rate

For each booking this month: hours = (Booking.createdAt - Sample.receivedDate). On-time if hours $\leq$ 2. Count on-time / total valid x 100. Only bookings with 0-720 hour range counted.

Avg Time

Average of (Booking.createdAt - Sample.receivedDate) in hours for all bookings this month. This is the Booking TAT: sample received at lab to booking completed.

All stats refresh every 60 seconds.

KPIs (6 Metrics)

KPI	Target	What It Measures
Booked / Day	25	Volume of bookings today
Avg Booking Time	2 hrs	TAT: sample received to booking created
On-Time Rate	95%	% booked within 2 hours of receipt
Amendment Rate	5%	% of bookings needing corrections (inverse)
Barcode Same Day	100%	% of barcodes generated same day
Pending Queue	0	Lab-wide unbooked samples (inverse)

Booking TAT Definition

Start: Sample.receivedDate (when sample physically arrived at the lab)

End: Booking.createdAt (when booking person completed the booking)

On-time threshold: 2 hours or less between these two timestamps.

Only bookings where time difference is 0-720 hours are counted (filters bad data).

KPI Progress Formulas

- Normal: $\min(100, \text{actual} / \text{target} \times 100)$
- Inverse (target=0): actual is 0 = 100%, else $\max(0, 100 - \text{actual} \times 10)$
- Inverse (target>0): $\min(100, \text{target} / \text{actual} \times 100)$

KRAs (5 Weighted Areas)

KRA	Target	Weight	Calculation
Booking Accuracy	99%	30%	(1 - amended / total) x 100
Booking Turnaround	95%	25%	% booked within 2 hrs of receipt
Daily Volume	100%	20%	total month / (working days x 25) x 100
Amendment Rate	<=5%	15%	amended / total x 100 (inverse)
Client Compliance	100%	10%	% with client name+email+phone+address

Client Data Compliance (Fixed)

Previously hardcoded at 97%. Now auto-computed by checking each booking:

- Client has name filled?
- Client has email filled?
- Client has phone filled?
- Client has address filled?

If all four present = compliant. Percentage = compliant / total x 100.

Overall KRA Formula

Normal: score = min(100, (actual / target) x 100)

Inverse: if actual <= target, score = 100. Else: max(0, 100 - ((actual - target) / target) x 100)

Overall = SUM(score x weight / 100) for all 5 KRAs

Example:

- Accuracy: 97.8% / 99% target = score 98.8 x 0.30 = 29.6
- Turnaround: 92.1% / 95% target = score 96.9 x 0.25 = 24.2
- Volume: 76% x 0.20 = 15.2
- Amendment: 3.8% (<=5% target) = score 100 x 0.15 = 15.0
- Compliance: 98.5% / 100% = score 98.5 x 0.10 = 9.9

Overall = 29.6 + 24.2 + 15.2 + 15.0 + 9.9 = 93.9%

Booking Queue

Samples in "received" status, sorted by receivedDate ascending. Limited to 50.

Wait Time	Color	Meaning
<= 2 hours	Green	Within target
2 - 4 hours	Yellow	Getting slow
> 4 hours	Red	Needs attention

Auto-Refresh Schedule

Data	Refresh	Stale Time	Reason
Booking Queue	30 seconds	15s	New samples arrive frequently
Stats, Bookings	60 seconds	30s	Work changes during the day
KRAs	5 minutes	60s	Monthly metrics change slowly

Data Sources

What	Database Table
Booking counts, amounts, amendments	Booking
Sample receivedDate, status, priority	Sample
Client name, email, phone, address	Client
Test count per booking	BookingTest

Pending Items

Item	Current	Planned
Barcode Tracking	Approximated	Track actual barcode timestamp
Quotation Module	Not on dashboard	Add quotation status
PO Consumption	Not on dashboard	Add PO tracking widget
Cancellation Rate	Calculated, not shown	Add to KPI sidebar